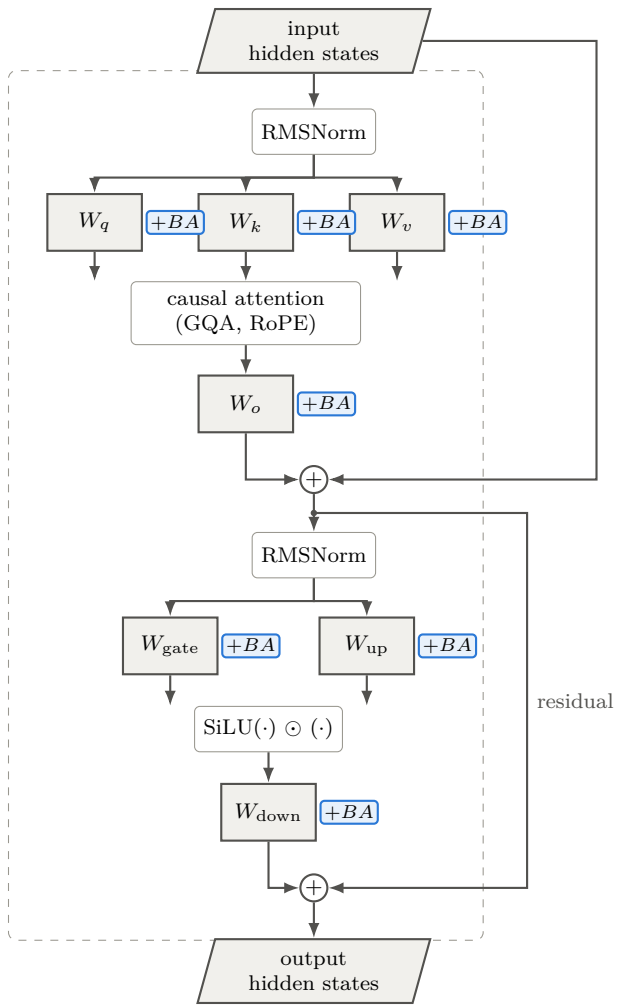



**one Qwen2.5 transformer block, repeated 24 times (0.5B) or 36 (3B)**

$W$ — base model weights, frozen at 8 bits

$+BA$ — trainable LoRA adapter (rank 16, scale 2)

trainable parameters per block:

- 0.5B: 366,592
- 3B: 831,488

in total:

- 0.5B: 8.8 million (1.8 % of the model)
- 3B: 29.9 million (1.0 % of the model)

Embeddings and output head (shared) stay frozen, without adapters. The arrows on the right are the residual connections.